

ALLIANCE KCSE 2025 PREDICTION



POSSIBLE KCSE EXAM



AGRICULTURE

443/1

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your name and index number in the spaces provided above.
- The paper consists of **three** sections A , B and C.
- Answer all questions in section A and B in the spaces provided
- Answer **any two** questions from section C in the spaces provided

FOR EXAMINER'S USE ONLY

Section	Question	Maximum score	Candidate's score
A	1-16	30	
B	17	20	
	18		
	19		
	20		
C	21-23	40	
	TOTAL	90	

SECTION A (30 marks)

1. Give two reasons why agriculture is looked as an artistic practice
(2marks)

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2. Outline two disadvantages of communal land tenure system
(1mark)

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3. Give three different types of soil water
(1 ½ marks)

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4. Give two ways of utilization of napier grass as a forage crop
(1mark)

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5. Give two signs of crop infestation by root nodes nematodes
(1mark)

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6. Name two chemicals used in water treatment

(1mark)

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7. a) A farmer was advised to apply 210 kg of calcium of ammonium nitrate

(CAN 21-0-0) per hectare. Calculate the amount of nitrogen required

(2 marks)

b) Different between fertilizer grade and fertilizer ratio

(2marks)

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8. Give four biotic factors that influence agriculture

(2marks)

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9. a) Give reasons why land should be prepared early in readiness for planting

(2marks)

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(b) State reasons for deep ploughing during land preparation

(2marks)

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10. List two methods of land reclamation

(2marks)

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11. A form four student planted maize for her KCSE agriculture project. She use a spacing of 75cmx25cm; her plot was 4 x3m.

(i) What does 75cm stand for in this statement

(1mark)

(ii) Calculate the plant population showing your working

(3marks)

12. Name two pests with piercing and sucking mouth parts

(1mark)

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13. a) Explain the form changing the cycle in coffee growing
(1mark)

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b) State two importance of tissue culture in crop production
(1mark)

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14. Why are leguminous plants preferred for green manure?
(1mark)

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15. i) What is opportunity cost
(1mark)

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ii) A farmer has an option of growing either wheat or maize in her one hectare of land. Wheat gives a return of Kshs. 20, 000 while maize fetches Kshs. 35, 000 in the market. If the farmer chooses to grow wheat, what will be the opportunity cost? (1mark)

16. State three control measures of maize streak disease
(1.5marks)

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SECTION B (20 MARKS)

17. The table below shows output of maize in response to increase in D.A.P fertilizers on one hectare of land.

Fixed input land (in ha)	Variable input D.A.P in 30kg-bag	Total product maize yield in 90-kb bag	Average product (AP) maize in 90kg bag	Marginal product (M.P) in 90kg bag
1	0	2	-	-
1	1	5	-	-
1	2	14	-	-
1	3	21	-	-
1	4	26	-	-

a) Fill in the table for average product (A.P) and marginal products (M.P) (5marks)

18. The diagrams below illustrate irish potato seed preparation before planting. Study it carefully and answer the questions that follow.



a) Name the practice used in preparing the seed potato above before planting. (1mark)

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b) Describe the procedure followed in preparing the seed potatoes for planting. (3marks)

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c) Give one reason for carrying out the practice named above. (1mark)

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19. The photograph below shows an irish potato plant attacked by a disease. Study it and answer the questions that follow.



a) Identify the disease represented by the photograph. (1mark)

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b) Name the causative agent of the diseases. (1mark)

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c) Give **one** other crop that can be attacked by the same disease.

(1mark)

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d) List **two** control measures of the disease.

(2marks)

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20. The photographs below show common weeds C and D in pasture land. Study them carefully and answer the questions that follow.



C

D

a) Identify weeds C and D.

(2marks)

C.....D.....

b) Classify weed D according to plant morphology.

(1mark)

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- c) State the major problem posed by each of the weeds above in pasture land. (2marks)

C

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D

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SECTION C (40MARKS)

Answer Any TWO questions in this section

21. (a) Explain **FIVE** cultural methods of pest control (10marks)
(b) Outline **FIVE** factors considered in timely planting of annual crops (10marks)
22. (a) Outline the process of land Adjudication (5marks)
(b) Discuss **five** reasons for carrying out minimum tillage. (5 marks)
(c) Explain **5** ways in which soil losses fertility. (10 marks)
23. (a) Describe **seven** field management practices in tomato production. (7marks)
b. Outline **five** factors that determine water requirements in an animal's body. (5 marks)
c. Describe the transplanting of tree seedlings. (8 marks)

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